2.07 Sequences and Series: Arithmetic Sequences

Worked Solutions 1–52

1 a)

$a = 3$, $d = 8 - 3 = 5$. $u_n = 3 + (n-1) \times 5$.

1 b)

$a = 25$, $d = 35 - 25 = 10$. $u_n = 25 + (n-1) \times 10$.

1 c)

$a = 12$, $d = 6 - 12 = -6$. $u_n = 12 + (n-1) \times -6$.

1 d)

$a = 150$, $d = 135 - 150 = -15$. $u_n = 150 + (n-1) \times -15$.

1 e)

$a = -44$, $d = -41 - (-44) = 3$. $u_n = -44 + (n-1) \times 3$.

1 f)

$a = -100$, $d = -121 - (-100) = -21$. $u_n = -100 + (n-1) \times -21$.

1 g)

$a = \frac{5}{6}$, $d = \frac{31}{30} - \frac{5}{6} = \frac{1}{5}$. $u_n = \frac{5}{6} + (n-1) \times \frac{1}{5}$.

1 h)

$a = \frac{11}{4}$, $d = -\frac{7}{4} - \frac{11}{4} = -\frac{9}{2}$. $u_n = \frac{11}{4} + (n-1) \times -\frac{9}{2}$.

2)

$a = 9$, $d = 12 - 9 = 3$. $u_n = 9 + (n-1) \times 3$. $u_{500} = 9 + (500-1) \times 3 = 1506$.

3)

$a = \frac{3}{2}$, $d = \frac{11}{6} - \frac{3}{2} = \frac{1}{3}$. $u_n = \frac{3}{2} + (n-1) \times \frac{1}{3}$. $u_{240} = \frac{3}{2} + (240-1) \times \frac{1}{3} = \frac{487}{6}$.

4)

$a = -50$, $d = -45 - (-50) = 5$. $u_n = -50 + (n-1) \times 5$. $765 = -50 + (n-1) \times 5 \Rightarrow n = 164$. So there are 164 terms in the sequence.

5)

$a = 31.3$, $d = 31.6 - 31.3 = 0.3$. $u_n = 31.3 + (n-1) \times 0.3$. $117.7 = 31.3 + (n-1) \times 0.3 \Rightarrow n = 289$.

6)

$a = 58.2$, $d = 57.5 - 58.2 = -0.7$. $u_n = 58.2 + (n-1) \times -0.7$. $-25.1 = 58.2 + (n-1) \times -0.7 \Rightarrow n = 120$.

7)

$u_3 = 22 \Rightarrow 22 = a + (3-1) \times d \Rightarrow a + 2d = 22$ (1). $u_7 = 54 \Rightarrow 54 = a + (7-1) \times d \Rightarrow a + 6d = 54$ (2). $\therefore a = 6$, $d = 8$.

8)

$u_5 = 38 \Rightarrow 38 = a + (5-1) \times d \Rightarrow a + 4d = 38$ (1). $u_{10} = 50.5 \Rightarrow 50.5 = a + (10-1) \times d \Rightarrow a + 9d = 50.5$ (2). $\therefore a = 28$, $d = \frac{5}{2}$.

9)

$u_2 = 71 \Rightarrow 71 = a + (2-1) \times d \Rightarrow a + d = 71$ (1). $u_9 = 631 \Rightarrow 631 = a + (9-1) \times d \Rightarrow a + 8d = 631$ (2). $\therefore a = -9$, $d = 80$.

10)

$u_6 = 20 \Rightarrow 20 = a + (6-1) \times d \Rightarrow a + 5d = 20$ (1). $u_{12} = 38 \Rightarrow 38 = a + (12-1) \times d \Rightarrow a + 11d = 38$ (2). $\therefore a = 5$, $d = 3$. $5 + (n-1) \times 3 > 100 \Rightarrow n > \frac{98}{3} \approx 32.7$. So the 33rd term, $u_{33} = 101$, will be the first to exceed 100.

11)

$u_3 = 2 + \sqrt{3} \Rightarrow 2 + \sqrt{3} = a + (3-1) \times d \Rightarrow a + 2d = 2 + \sqrt{3}$ (1). $u_9 = 3 + \sqrt{5} \Rightarrow 3 + \sqrt{5} = a + (9-1) \times d \Rightarrow a + 8d = 3 + \sqrt{5}$ (2). (1) : $a = 2 + \sqrt{3} - 2d$. (2) : $a = 3 + \sqrt{5} - 8d$. $2 + \sqrt{3} - 2d = 3 + \sqrt{5} - 8d \Rightarrow 6d = 1 - \sqrt{3} + \sqrt{5}$. $\therefore d = \frac{1}{6} - \frac{1}{6}\sqrt{3} + \frac{1}{6}\sqrt{5}$.

12)

$u_2 = 15 \Rightarrow 15 = a + (2-1) \times d \Rightarrow a + d = 15$ (1). $u_7 = 55 \Rightarrow 55 = a + (7-1) \times d \Rightarrow a + 6d = 55$ (2). $\therefore a = 7$, $d = 8$. $u_3 = 7 + (3-1) \times 8 = 23$ cm.

13)

$$d = \ln \tfrac{1}{3}k - \ln \tfrac{1}{4}k = \ln\left(\frac{\frac{1}{3}k}{\frac{1}{4}k}\right) = \ln \frac{4}{3},$$
$$d = \ln p - \ln \tfrac{1}{3}k \Rightarrow \ln p - \ln \tfrac{1}{3}k = \ln \frac{4}{3},$$
$$\ln p = \ln \frac{4}{3} + \ln \tfrac{1}{3}k = \ln\left(\frac{4}{3} \times \frac{1}{3}k\right) = \ln \frac{4}{9}k.$$

$\therefore p = \frac{4}{9}k$.

14)

$d = 5k - 6k^2$, $d = (21k - 35) - 5k$. $5k - 6k^2 = (21k - 35) - 5k \Rightarrow 6k^2 + 11k - 35 = 0 \Rightarrow k = -\frac{7}{2}$ or $k = \frac{5}{3}$.

15)

$$d = 20k - \frac{2}{k}, \qquad d = -(5k + 13) - 20k,$$
$$20k - \frac{2}{k} = -(5k + 13) - 20k = -25k - 13,$$

$$20k^2 - 2 = -25k^2 - 13k,$$
$$45k^2 + 13k - 2 = 0 \Rightarrow k = -\frac{2}{5} \text{ or } k = \frac{1}{9}.$$

16)

$$d = 20 - 16e^k, \qquad d = 16e^{-k} - 20,$$
$$20 - 16e^k = 16e^{-k} - 20 \Rightarrow 40 - 16e^k = 16e^{-k},$$
$$40e^k - 16e^{2k} = 16 \Rightarrow 16e^{2k} - 40e^k + 16 = 0,$$
$$2e^{2k} - 5e^k + 2 = 0 \Rightarrow (2e^k - 1)(e^k - 2) = 0.$$

$2e^k - 1 = 0 \Rightarrow e^k = \frac{1}{2} \Rightarrow k = \ln \frac{1}{2} = -\ln 2$. $e^k - 2 = 0 \Rightarrow e^k = 2 \Rightarrow k = \ln 2$. $\therefore k = \ln \frac{1}{2}$ or $k = \ln 2$.

17)

$$d = \frac{200}{2^k} - 28, \qquad d = \left(\frac{240}{2^k} - 2^k\right) - \frac{200}{2^k},$$
$$\frac{200}{2^k} - 28 = \left(\frac{240}{2^k} - 2^k\right) - \frac{200}{2^k} = \frac{40}{2^k} - 2^k,$$
$$\frac{160}{2^k} + 2^k - 28 = 0 \Rightarrow 160 + 2^{2k} - 28 \times 2^k = 0,$$
$$2^{2k} - 28 \times 2^k + 160 = 0 \Rightarrow (2^k - 8)(2^k - 20) = 0.$$

$2^k - 8 = 0 \Rightarrow 2^k = 8 \Rightarrow k = 3$. $2^k - 20 = 0 \Rightarrow 2^k = 20 \Rightarrow k = \log_2 20$. $\therefore k = 3$ or $k = \log_2 20$.

18 a)

$d = (13-4k)-(32+3k) = -19-7k$, $d = (78+10k)-(13-4k) = 65+14k$. $-19-7k = 65+14k \Rightarrow -84 = 21k \Rightarrow k = -4$.

18 b)

$a = 32+3\times-4 = 20$, $d = -19-7\times-4 = 9$. $u_n = 20+(n-1)\times 9$. $u_{100} = 20+(100-1)\times 9 = 911$.

19)

$a = 3$, $d = 11-3 = 8$. $u_n = 3+(n-1)\times 8$. $155 = 3+(n-1)\times 8 \Rightarrow n = 20$. $S_{20} = \frac{1}{2}\times 20\times(3+155) = 1580$.

20)

$a = 14$, $d = 20 - 14 = 6$. $u_n = 14 + (n-1) \times 6$. $1238 = 14 + (n-1) \times 6 \Rightarrow n = 205$. $S_{205} = \frac{1}{2} \times 205 \times (14 + 1238) = 128330$.

21)

$a = -11$, $d = -20-(-11) = -9$. $u_n = -11+(n-1)\times -9$. $-542 = -11+(n-1)\times -9 \Rightarrow n = 60$. $S_{60} = \frac{1}{2} \times 60 \times (-11 + (-542)) = -16590$.

22)

$a = \frac{20}{3}$, $d = \frac{35}{6} - \frac{20}{3} = -\frac{5}{6}$. $u_n = \frac{20}{3} + (n-1) \times -\frac{5}{6}$. $-60 = \frac{20}{3} + (n-1) \times -\frac{5}{6} \Rightarrow n = 81$. $S_{81} = \frac{1}{2} \times 81 \times (\frac{20}{3} + (-60)) = -2160$.

23 a)

$a = 6 \times 7 - \frac{1}{2} = \frac{83}{2}$, $d = 6$.

23 b)

$40 - 7 + 1 = 34$ terms.

24)

$a = u_1 = 7 - 10 \times 1 = -3$, $l = u_{40} = 7 - 10 \times 40 = -393$. $\Sigma = \frac{1}{2} \times 40 \times (-3 + (-393)) = -7920$.

25)

$a = u_{10} = -3 + \frac{10}{3} = \frac{1}{3}$, $l = u_{30} = -3 + \frac{30}{3} = 7$. $n = 30 - 10 + 1 = 21$. $\Sigma = \frac{1}{2} \times 21 \times (\frac{1}{3} + 7) = 77$.

26)

$a = 25$, $d = 26.5 - 25 = 1.5$, $n = 150$. $S_{150} = \frac{1}{2} \times 150 \times (2 \times 25 + (150-1) \times 1.5) = 20512.5$.

27)

$a = \frac{1}{5}$, $d = \frac{7}{5} - \frac{1}{5} = \frac{6}{5}$, $n = 80$. $S_{80} = \frac{1}{2} \times 80 \times (2 \times \frac{1}{5} + (80-1) \times \frac{6}{5}) = 3808$.

28)

$a = 100$, $d = 99.2 - 100 = -0.8$, $n = 185$. $S_{185} = \frac{1}{2} \times 185 \times (2 \times 100 + (185-1) \times -0.8) = 4884$.

29)

$a = 10.3$, $d = 10 - 10.3 = -0.3$, $n = 390$. $S_{390} = \frac{1}{2} \times 390 \times (2 \times 10.3 + (390-1) \times -0.3) = -18739.5$.

30)

$a = 35$, $d = 65 - 35 = 30$, $n = 18$. $S_{18} = \frac{1}{2} \times 18 \times (2 \times 35 + (18-1) \times 30) = 5220$. So there is £5220 in the bank account the day after Jaime's 18th birthday.

31)

$a = 104$, $d = 101 - 104 = -3$. $u_n = 104 + (n-1) \times -3$. Once the terms become negative, the value of S_n will decrease, so first find the last positive u_n. $0 = 104 + (k-1) \times -3 \Rightarrow k = \frac{107}{3} \approx 35.7$. So u_{35} is the last positive term in the sequence. $u_{35} = 104 + (35-1) \times -3 = 2$. $\therefore S_{35} = \frac{1}{2} \times 35 \times (104+2) = 1855$.

32)

$a = \frac{6 \times 1}{5} + \frac{2}{3} = \frac{28}{15}$, $d = \frac{6}{5}$.

$$
\begin{aligned}
200 &= \frac{1}{2} \times n \times \left(2 \times \frac{28}{15} + (n-1) \times \frac{6}{5}\right), \\
400 &= \frac{56}{15}n + \frac{6}{5}(n^2 - n), \\
400 &= \frac{56}{15}n + \frac{6}{5}n^2 - \frac{6}{5}n,
\end{aligned}
$$

$$\frac{6}{5}n^2 + \frac{38}{15}n - 400 = 0,$$
$$n = 17.232\ldots \text{ or } n = -19.343\ldots.$$

n cannot be negative, so $n = 17.232\ldots$. $\therefore k = 17$.

33)

$a = 4.1 \times 1 - 3 = 1.1$, $d = 4.1$.

$$3000 = \frac{1}{2} \times n \times (2 \times 1.1 + (n-1) \times 4.1),$$
$$6000 = 2.2n + 4.1(n^2 - n),$$
$$6000 = 2.2n + 4.1n^2 - 4.1n,$$
$$4.1n^2 - 1.9n - 6000 = 0,$$
$$n = 38.487\ldots \text{ or } n = -38.023\ldots.$$

n cannot be negative, so $n = 38.487\ldots$. $\therefore k = 38$.

34 a)

$a = 10.5$, $d = 10.6 - 10.5 = 0.1$.

$$900 = \frac{1}{2} \times n \times (2 \times 10.5 + (n-1) \times 0.1),$$
$$1800 = 21n + 0.1(n^2 - n),$$
$$1800 = 21n + 0.1n^2 - 0.1n,$$
$$0.1n^2 + 20.9n - 1800 = 0,$$
$$n = 65.559\ldots \text{ or } n = -274.559\ldots.$$

n cannot be negative, so $n = 65.559\ldots$. So the arithmetic series has 65 terms, so there are 65 payments before the final payment to make up the final amount to £900. So 66 weeks.

34 b)

$S_{65} = \frac{1}{2} \times 65 \times (2 \times 10.5 + (65-1) \times 0.1) = 890.5$. So at the end of the 65th week, Aiden has paid £890.50. So in the final week he pays the final £9.50.

35)

$u_3 = 46 \Rightarrow 46 = a + (3-1) \times d \Rightarrow a + 2d = 46$ (1). $u_{15} = 222 \Rightarrow 222 = a + (15-1) \times d \Rightarrow a + 14d = 222$ (2). $\therefore a = \frac{50}{3}$, $d = \frac{44}{3}$. $S_{30} = \frac{1}{2} \times 30 \times (2 \times \frac{50}{3} + (30-1) \times \frac{44}{3}) = 6880$.

36)

$a = 3$, $S_{100} = 12$. $12 = \frac{1}{2} \times 100 \times (2 \times 3 + (100-1) \times d)$. $0.24 = 6 + 99d \Rightarrow d = -\frac{16}{275}$.

37)

$$d = k^2(k-4) - 8k, \qquad d = 10 - k^2(k-4),$$
$$k^2(k-4) - 8k = 10 - k^2(k-4),$$
$$2k^2(k-4) - 8k - 10 = 0,$$
$$2k^3 - 8k^2 - 8k - 10 = 0 \Rightarrow k = 5.$$

$a = 8 \times 5 = 40$, $d = 5^2 \times (5-4) - 8 \times 5 = -15$.

$$\begin{aligned} -53900 &= \frac{1}{2} \times n \times (2 \times 40 + (n-1) \times -15), \\ -107800 &= 80n - 15(n^2 - n), \\ -107800 &= 80n - 15n^2 + 15n, \\ 15n^2 - 95n - 107800 = 0 \Rightarrow n &= -\frac{245}{3} \text{ or } n = 88. \end{aligned}$$

n must be a positive integer, so $n = 88$.

38 a)

$S_{15} = -465$. $-465 = \frac{1}{2} \times 15 \times (2a + (15-1) \times d)$. $-62 = 2a + 14d \Rightarrow a + 7d = -31$ (1).

38 b)

$u_{50} = -283 \Rightarrow -283 = a + (50-1) \times d \Rightarrow a + 49d = -283$ (2). $\therefore a = 11$, $d = -6$.

39 a)

$S_{10} = 410$. $410 = \frac{1}{2} \times 10 \times (2a + (10-1) \times d)$. $410 = 10a + 45d \Rightarrow 2a + 9d = 82$ (1).

39 b)

$S_{20} = 1620$. $1620 = \frac{1}{2} \times 20 \times (2a + (20-1) \times d)$. $1620 = 20a + 190d \Rightarrow 2a + 19d = 162$ (2). $\therefore a = 5$, $d = 8$.

39 c)

$S_{30} = \frac{1}{2} \times 30 \times (2 \times 5 + (30-1) \times 8) = 3630$.

40 a)

$$\begin{aligned} S_{490} &= 1489 \times S_{10}, \\ \frac{1}{2} \times 490 \times (2a + (490-1) \times d) &= 1489 \times \frac{1}{2} \times 10 \times (2a + (10-1) \times d), \\ 490(2a + 489d) &= 14890(2a + 9d), \\ 980a + 239610d &= 29780a + 134010d, \\ 105600d &= 28800a \Rightarrow 3a = 11d \text{ (1)}. \end{aligned}$$

40 b)

$u_{10} = 38 \Rightarrow 38 = a + (10-1) \times d \Rightarrow a + 9d = 38$ (2). $\therefore a = 11$, $d = 3$.

41 a)

$$\begin{aligned} S_{220} &= 211 \times S_{20}, \\ \frac{1}{2} \times 220 \times (2a + (220-1) \times d) &= 211 \times \frac{1}{2} \times 20 \times (2a + (20-1) \times d), \\ 220(2a + 219d) &= 4220(2a + 19d), \\ 440a + 48180d &= 8440a + 80180d, \\ 0 &= 8000a + 32000d \Rightarrow a + 4d = 0 \text{ (1)}. \end{aligned}$$

41 b)

$u_{105} = -200 \Rightarrow -200 = a + (105-1) \times d \Rightarrow a + 104d = -200$ (2). $\therefore a = 8$, $d = -2$.

42 a)

$$\begin{aligned}
S_{425} &= 221 \times (S_5)^2, \\
\frac{1}{2} \times 425 \times (2a + (425-1) \times d) &= 221 \times \left(\frac{1}{2} \times 5 \times (2a + (5-1) \times d)\right)^2, \\
\frac{425}{2}(2a + 424d) &= 221 \left(\frac{5}{2}(2a + 4d)\right)^2, \\
425a + 90100d &= 221(5a + 10d)^2, \\
425a + 90100d &= 221(25a^2 + 100ad + 100d^2), \\
425a + 90100d &= 5525a^2 + 22100ad + 22100d^2, \\
17a + 3604d &= 221a^2 + 884ad + 884d^2, \\
221a^2 - 17a + 884d^2 - 3604d + 884ad &= 0.
\end{aligned}$$

42 b)

$u_2 = 4 \Rightarrow 4 = a + (2-1) \times d \Rightarrow a + d = 4 \Rightarrow d = 4 - a$.

$$\begin{aligned}
221a^2 - 17a + 884(4-a)^2 - 3604(4-a) + 884a(4-a) &= 0, \\
221a^2 - 17a + 884(16 - 8a + a^2) - 14416 + 3604a + 3536a - 884a^2 &= 0, \\
221a^2 - 17a + 14144 - 7072a + 884a^2 - 14416 + 3604a + 3536a - 884a^2 &= 0, \\
221a^2 + 51a - 272 &= 0, \\
17a^2 + 3a - 16 = 0 \Rightarrow a = -\frac{16}{13} \text{ or } a &= 1.
\end{aligned}$$

$a \neq -\frac{16}{13}$ as a is a positive integer. So $a = 1$. $\therefore d = 4 - 1 = 3$.

43)

$$d = -\frac{1}{k} - \frac{1}{k-4}, \qquad d = \frac{1}{k+4} - \left(-\frac{1}{k}\right),$$

$$\begin{aligned}
-\frac{1}{k} - \frac{1}{k-4} &= \frac{1}{k+4} + \frac{1}{k}, \\
-(k-4)(k+4) - k(k+4) &= k(k-4) + (k-4)(k+4), \\
-k^2 + 16 - k^2 - 4k &= k^2 - 4k + k^2 - 16, \\
-2k^2 - 4k + 16 &= 2k^2 - 4k - 16, \\
4k^2 - 32 = 0 \Rightarrow k^2 - 8 &= 0, \\
k = \pm\sqrt{8} &= \pm 2\sqrt{2}.
\end{aligned}$$

k is a positive constant so $k = 2\sqrt{2}$. $a = \frac{1}{2\sqrt{2}-4}$, $d = -\frac{1}{2\sqrt{2}} - \frac{1}{2\sqrt{2}-4}$.

$$\begin{aligned}
S_{52} &= \frac{1}{2} \times 52 \times \left(2 \times \frac{1}{2\sqrt{2}-4} + (52-1) \times \left(-\frac{1}{2\sqrt{2}} - \frac{1}{2\sqrt{2}-4}\right)\right) \\
&= 26 \left(\frac{2}{2\sqrt{2}-4} - \left(\frac{51}{2\sqrt{2}} + \frac{51}{2\sqrt{2}-4}\right)\right)
\end{aligned}$$

$$= 26\left(-\frac{49}{2\sqrt{2}-4} - \frac{51}{2\sqrt{2}}\right) = -13\left(\frac{49}{\sqrt{2}-2} + \frac{51}{\sqrt{2}}\right)$$
$$= -13\left(\frac{49}{\sqrt{2}-2} \times \frac{\sqrt{2}+2}{\sqrt{2}+2} + \frac{51}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}}\right)$$
$$= -13\left(\frac{49\sqrt{2}+98}{2-4} + \frac{51\sqrt{2}}{2}\right) = -13\left(\frac{49\sqrt{2}+98}{-2} + \frac{51\sqrt{2}}{2}\right)$$
$$= -\frac{13}{2}(51\sqrt{2} - 49\sqrt{2} - 98)$$
$$= -\frac{13}{2}(2\sqrt{2} - 98) = 637 - 13\sqrt{2}.$$

44)

$a = 8$, $d = 11 - 8 = 3$.

$$1000 < \frac{1}{2} \times n \times (2 \times 8 + (n-1) \times 3) < 2000,$$
$$2000 < n(16 + 3n - 3) < 4000,$$
$$2000 < n(13 + 3n) < 4000,$$
$$2000 < 13n + 3n^2 < 4000.$$

$3n^2 + 13n - 2000 > 0 \Rightarrow n < -28.077\ldots$ or $n > 23.743\ldots$. As n is a positive integer, we must have $n \geq 24$. $3n^2 + 13n - 4000 < 0 \Rightarrow -38.745 < n < 34.412\ldots$. As n is a positive integer, we must have $1 \leq n \leq 34$. So the two regions are both true when $24 \leq n \leq 34$.

45)

$a = p \times 1 + q \Rightarrow a = p + q$, $d = p$, $S_{100} = 100$.

$$100 = \frac{1}{2} \times 100 \times (2(p+q) + (100-1) \times p),$$
$$2 = 2p + 2q + 99p \Rightarrow 101p + 2q = 2 \ (1).$$

$$u_{51} + u_{53} + u_{55} = u_{59},$$
$$\Rightarrow ((p+q) + (51-1) \times p) + ((p+q) + (53-1) \times p)$$
$$+ ((p+q) + (55-1) \times p) = ((p+q) + (59-1) \times p),$$
$$(p + q + 50p) + (p + q + 52p) + (p + q + 54p)$$
$$= (p + q + 58p),$$
$$51p + q + 53p + q + 55p + q = 59p + q,$$
$$100p + 2q = 0 \ (2).$$

$\therefore p = 2$, $q = -100$.

46)

$a = 3k+2$, $d = 6$, $l = 9k-1$. $9k-1 = (3k+2) + (n-1) \times 6 \Rightarrow 6k - 3 = 6n - 6 \Rightarrow n = \frac{6k+3}{6} = k + \frac{1}{2}$.

47)

$$\begin{aligned} A_k &= B_k, \\ \frac{1}{2} \times k \times (2 \times 3 + (k-1) \times 8) &= \frac{1}{2} \times k \times (2 \times -52 + (k-1) \times 18), \\ 6 + 8(k-1) &= -104 + 18(k-1), \\ 110 &= 10(k-1) \Rightarrow k - 1 = 11 \Rightarrow k = 12. \end{aligned}$$

48)

$$\begin{aligned} C_k &= D_k, \\ \frac{1}{2} \times k \times (2 \times 1500 + (k-1) \times -9) &= \frac{1}{2} \times k \times (2 \times 6 + (k-1) \times 3), \\ 3000 - 9(k-1) &= 12 + 3(k-1), \\ 2988 &= 12(k-1) \Rightarrow k - 1 = 249 \Rightarrow k = 250. \end{aligned}$$

49)

$$\begin{aligned} S &= a + (a+d) + (a+2d) + \cdots + (a-2l) + (a-l) + l, \\ S &= l + (a-l) + (a-2l) + \cdots + (a+2d) + (a+d) + a, \\ 2S &= (a+l) + (a+l) + (a+l) + \cdots + (a+l) + (a+l) + (a+l), \\ 2S &= n(a+l) \Rightarrow S = \frac{1}{2} n(a+l). \end{aligned}$$

Using $l = a + (n-1)d$: $S = \frac{1}{2} n(a + (a + (n-1)d)) \Rightarrow S = \frac{1}{2} n(2a + (n-1)d)$.

50)

$a = 100$, $d = 2$, $l = 500$. $500 = 100 + (n-1) \times 2 \Rightarrow n = 201$. $S = \frac{1}{2} \times 201 \times (2 \times 100 + (201-1) \times 2) = \frac{201}{2}(200 + 400) = 60300$.

51)

$a = 5$, $d = 2$, $l = 555$. $555 = 5 + (n-1) \times 2 \Rightarrow n = 276$. $S = \frac{1}{2} \times 276 \times (2 \times 5 + (276-1) \times 2) = 138(10 + 550) = 77280$.

52)

$$\begin{aligned} v_1 &= \frac{u_2}{u_1} = \frac{2p-1}{p}, \\ v_2 &= \frac{u_3}{u_2} = \frac{3p+2}{2p-1}, \\ v_3 &= \frac{u_4}{u_3} = \frac{15p + \frac{2}{3}}{3p+2}, \\ d &= v_2 - v_1 = \frac{3p+2}{2p-1} - \frac{2p-1}{p}, \end{aligned}$$

$$
\begin{aligned}
d = v_3 - v_2 = \frac{15p + \frac{2}{3}}{3p+2} - \frac{3p+2}{2p-1}, \\
\frac{3p+2}{2p-1} - \frac{2p-1}{p} = \frac{15p + \frac{2}{3}}{3p+2} - \frac{3p+2}{2p-1}, \\
\frac{p(3p+2) - (2p-1)^2}{p(2p-1)} = \frac{(15p + \frac{2}{3})(2p-1) - (3p+2)^2}{(3p+2)(2p-1)}, \\
\frac{3p^2 + 2p - 4p^2 + 4p - 1}{p(2p-1)} = \frac{30p^2 - \frac{41}{3}p - \frac{2}{3} - 9p^2 - 12p - 4}{(3p+2)(2p-1)}, \\
\frac{-p^2 + 6p - 1}{p(2p-1)} = \frac{21p^2 - \frac{77}{3}p - \frac{14}{3}}{(3p+2)(2p-1)}, \\
\frac{-3p^2 + 18p - 3}{p(2p-1)} = \frac{63p^2 - 77p - 14}{(3p+2)(2p-1)}, \\
(-3p^2 + 18p - 3)(3p+2)(2p-1) = (63p^2 - 77p - 14)(2p-1)p, \\
(-3p^2 + 18p - 3)(6p^2 + p - 2) = (63p^2 - 77p - 14)(2p^2 - p), \\
-18p^4 + 105p^3 + 6p^2 - 39p + 6 = 126p^4 - 217p^3 + 49p^2 + 14p, \\
-144p^4 + 322p^3 - 43p^2 - 53p + 6 = 0, \\
p = -\frac{3}{8} \text{ or } p = \frac{1}{9} \text{ or } p = \frac{1}{2} \text{ or } p = 2.
\end{aligned}
$$

As $p \geq 1$, we must have $p = 2$.

$$
\begin{aligned}
v_3 = \frac{15 \times 2 + \frac{2}{3}}{3 \times 2 + 2} = \frac{23}{6}, \\
d = \frac{3 \times 2 + 2}{2 \times 2 - 1} - \frac{2 \times 2 - 1}{2} = \frac{7}{6}, \\
u_4 = 15 \times 2 + \frac{2}{3} = \frac{92}{3}, \\
v_4 = \frac{u_5}{u_4} \text{ and } v_4 = v_3 + d, \\
v_3 + d = \frac{u_5}{u_4} \Rightarrow \frac{23}{6} + \frac{7}{6} = \frac{u_5}{92/3} \Rightarrow u_5 = \frac{460}{3}.
\end{aligned}
$$